

# PE-EC801A — Antennas and Propagation

Semester VIII • 3L:0T:0P • 3 Credits

## Module 5 Exam Notes

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## 1. Characteristics of Microstrip Antennas

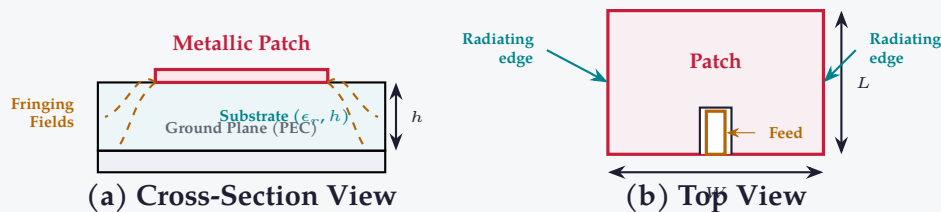
### Definition of Microstrip Patch Antenna

A **microstrip patch antenna** consists of a radiating metallic patch (usually rectangular or circular) etched on one side of a thin, grounded dielectric substrate (dielectric constant  $\epsilon_r$  and thickness  $h$ , where  $h \ll \lambda$ ).

### 1.1 Physical Structure and Radiation Mechanism

The metallic patch is usually made of copper or gold. The substrate thickness is typically  $0.003\lambda_0 \leq h \leq 0.05\lambda_0$ . The patch acts as a resonant cavity with open-circuited edges. Fringing fields at the edges of the patch (along the length) act as the source of radiation. These fields extend slightly outside the physical boundaries, meaning the patch behaves electrically as if it were slightly larger than its physical dimensions.

#### Rectangular Microstrip Patch Antenna (Cross-Section and Top View)



#### Advantages and Disadvantages:

| Advantages                                         | Disadvantages                                                    |
|----------------------------------------------------|------------------------------------------------------------------|
| Low profile, lightweight, conformable to surfaces. | Extremely narrow bandwidth (typically 1%–5%).                    |
| Low fabrication cost (easy to print on PCBs).      | Low radiation efficiency (due to dielectric and surface losses). |
| Supports dual-frequency and dual-polarization.     | Low power handling capability.                                   |

## 2. Feeding Methods

To transfer electromagnetic energy into the patch, several feed techniques are used:

**1. Microstrip Line Feed:** A microstrip conducting strip feed is directly connected to the edge of the patch.

- *Match method:* Inset cuts are made in the patch to match the high patch edge impedance to the feed line.
- *Pros/Cons:* Easy to fabricate; but causes spurious feed radiation.

**2. Coaxial Probe Feed:** The outer conductor of a coaxial cable is connected to the ground

plane, and the inner pin extends through the substrate and is soldered to the patch.

- *Match method*: Match is achieved by choosing the insertion point offset from the patch edges.
- *Pros/Cons*: Low spurious radiation; but requires drilling holes, making it harder to fabricate.

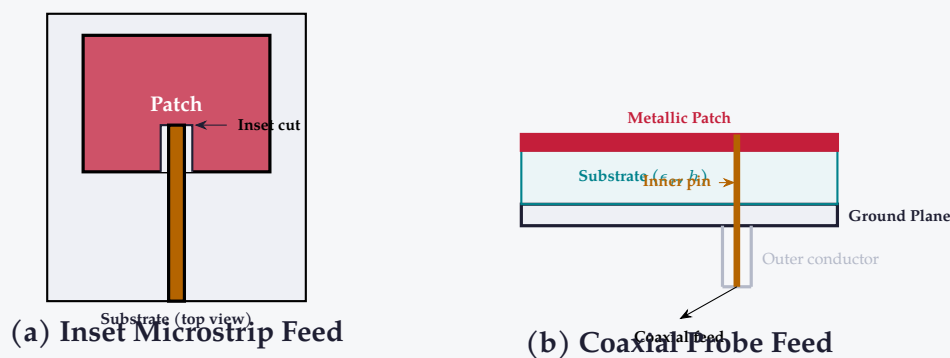
**3. Aperture-Coupled Feed:** The feed line is on a lower substrate, separated from the patch substrate by a common ground plane. Coupling occurs via a slot/aperture cut in the ground plane.

- *Pros/Cons*: Shields feed line spurious radiation, improves bandwidth; but requires multilayer fabrication.

**4. Proximity-Coupled Feed:** The feed line is sandwiched between two substrate layers, and the patch is on the top surface. Energy couples electromagnetically.

- *Pros/Cons*: Widest bandwidth (up to 13%), no contact needed; but complex to align.

#### Microstrip Line Feed vs. Coaxial Probe Feed



### 3. Methods of Analysis

- **Transmission Line Model**: Simplest model. It represents the patch as a microstrip transmission line of length  $L$  terminated by two radiating slots (modeled as parallel conductances  $G$  and susceptances  $B$ ). It yields good physical insight but is less accurate.
- **Cavity Model**: Represents the region between the patch and ground plane as a dielectric-loaded cavity with electric conductors on top/bottom, and artificial magnetic walls along the patch edges. It is more accurate and models higher-order modes.
- **Full-Wave Models**: Numerical techniques (such as the Method of Moments (MoM), Finite Element Method (FEM), or Finite Difference Time Domain (FDTD)). They are highly accurate for arbitrary shapes but are computationally demanding.

### 4. Design of Rectangular and Circular Patch Antennas

### 4.1 Rectangular Patch Antenna Design

The design parameters are: substrate height  $h$ , dielectric constant  $\epsilon_r$ , and resonant frequency  $f_r$ .

#### Rectangular Patch Design Calculations

1. **Width ( $W$ ):** To ensure high radiation efficiency, the width is:

$$W = \frac{v_0}{2f_r} \sqrt{\frac{2}{\epsilon_r + 1}}$$

where  $v_0 \approx 3 \times 10^8$  m/s.

2. **Effective Dielectric Constant ( $\epsilon_{reff}$ ):** Accounts for fringing fields:

$$\epsilon_{reff} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left[ 1 + 12 \frac{h}{W} \right]^{-1/2}$$

3. **Extended Patch Length ( $\Delta L$ ):** Due to fringing fields:

$$\Delta L = 0.412h \frac{(\epsilon_{reff} + 0.3) \left( \frac{W}{h} + 0.264 \right)}{(\epsilon_{reff} - 0.258) \left( \frac{W}{h} + 0.8 \right)}$$

4. **Physical Length ( $L$ ):**

$$L = L_{eff} - 2\Delta L = \frac{v_0}{2f_r \sqrt{\epsilon_{reff}}} - 2\Delta L$$

#### Worked Example: Rectangular Patch at 2.4 GHz

**Given:**  $f_r = 2.4$  GHz, FR-4 substrate:  $\epsilon_r = 4.4$ ,  $h = 1.6$  mm.

**Step 1 — Width:**

$$W = \frac{3 \times 10^8}{2 \times 2.4 \times 10^9} \sqrt{\frac{2}{4.4 + 1}} = 0.0625 \times 0.609 \approx 38.0 \text{ mm}$$

**Step 2 — Effective dielectric constant:**

$$\epsilon_{reff} = \frac{4.4 + 1}{2} + \frac{4.4 - 1}{2} \left[ 1 + 12 \frac{1.6}{38.0} \right]^{-1/2} = 2.7 + 1.7 \times 0.857 \approx 4.16$$

**Step 3 — Extension:**

$$\Delta L = 0.412 \times 1.6 \times \frac{(4.16 + 0.3)(38.0/1.6 + 0.264)}{(4.16 - 0.258)(38.0/1.6 + 0.8)} \approx 0.749 \text{ mm}$$

**Step 4 — Physical length:**

$$L = \frac{3 \times 10^8}{2 \times 2.4 \times 10^9 \times \sqrt{4.16}} - 2 \times 0.749 = 30.7 - 1.5 \approx 29.2 \text{ mm}$$

**Result:**  $W \approx 38$  mm,  $L \approx 29.2$  mm on a 1.6 mm FR-4 substrate.

## 4.2 Circular Patch Antenna Design

For a circular patch of radius  $a$  on a substrate of height  $h$ , the dominant resonant mode is  $TM_{110}^z$ .

### Resonant Modes and Radiation Mechanism

The circular patch behaves as a **resonant cavity** bounded by:

- **Top and bottom:** Perfect Electric Conductor (PEC) walls — the patch and ground plane.
- **Side boundary:** Perfect Magnetic Conductor (PMC) wall (fringing field approximation).

The resonant modes are designated  $TM_{nmp}^z$  where:

- $n$  = number of full-period variations in  $\phi$  (azimuthal direction)
- $m$  = number of zeros of  $J'_n(\cdot)$  in the radial direction ( $m = 1, 2, 3, \dots$ )
- $p$  = number of variations in  $z$  (always 0 for thin patches since  $h \ll \lambda$ )

The **dominant mode**  $TM_{110}^z$  has:

- One full-period azimuthal variation ( $n = 1$ ) and the first radial maximum ( $m = 1$ ).
- The constant 1.8412 is the first root of  $J'_1(\chi_{11}) = 0$  (derivative of first-order Bessel function), which defines the resonance condition.
- **Broadside radiation pattern** — maximum gain perpendicular to the patch surface ( $\theta = 0^\circ$ ).
- **Linear polarization** — the  $E$ -field across the radiating edge is linearly polarized.

### Higher-Order Modes

Other useful modes include:

- $TM_{010}^z$ : Omnidirectional (monopole-like) radiation; used for low-profile omnidirectional antennas.
- $TM_{210}^z$ : Figure-8 pattern; used for circular polarization when combined with  $TM_{110}^z$ .
- **Dual-Band Operation:** By stacking two circular patches of different radii on separate substrate layers, two resonant frequencies can be achieved independently.

### Circular Polarization from a Circular Patch

Circular polarization (CP) is an important feature achievable with circular patches:

- **Single feed CP:** A slight perturbation (small notch or truncation of the circular boundary) splits the degenerate  $TM_{110}^z$  mode into two orthogonal modes with a  $90^\circ$  phase separation. CP results when the two modes are excited with equal amplitude and  $90^\circ$  phase shift.
- **Two-feed CP:** Two coaxial probes placed at  $90^\circ$  apart on the patch perimeter, fed with equal amplitude and a  $90^\circ$  hybrid coupler, produce clean CP over a wider bandwidth.

## Fringing Fields and Effective Radius

The fringing fields at the patch edge extend the effective electrical radius beyond the physical radius:

$$a_e = a + \Delta a \quad \text{where} \quad \Delta a = a \left[ \sqrt{1 + \frac{2h}{\pi\epsilon_r a} \left( \ln \frac{\pi a}{2h} + 1.7726 \right)} - 1 \right]$$

Typically  $\Delta a/a \approx 1\%$ – $5\%$  for standard substrates. A higher  $\epsilon_r$  or thinner substrate reduces the fringing correction.

## Substrate Selection Tradeoffs

| Parameter            | High $\epsilon_r$ (e.g., 10) | Low $\epsilon_r$ (e.g., 2.2) |
|----------------------|------------------------------|------------------------------|
| Patch Size           | Smaller (compact)            | Larger                       |
| Bandwidth            | Narrower                     | Wider                        |
| Radiation Efficiency | Lower (more stored energy)   | Higher                       |
| Surface Wave Loss    | Higher                       | Lower                        |
| Typical Use          | Miniaturised antenna-on-chip | High-performance arrays      |

### Design Procedure:

1. Specify desired resonant frequency  $f_r$ , substrate  $\epsilon_r$  and thickness  $h$ .
2. Compute intermediate parameter  $F$  (an estimate of the effective radius assuming no fringing).
3. Solve for physical radius  $a$  by applying the fringing correction factor.
4. Compute effective radius  $a_e$ .
5. Verify by back-calculating  $(f_r)_{110}$  — should match the target within 1%.
6. Choose feed position (coaxial probe offset from centre) to match  $50\,\Omega$ .

**Typical example:** Circular patch at  $f_r = 2.4\,\text{GHz}$ , FR-4 substrate ( $\epsilon_r = 4.4$ ,  $h = 1.6\,\text{mm}$ ):  $F \approx 17.5\,\text{mm}$ ,  $a \approx 16.8\,\text{mm}$ ,  $a_e \approx 17.2\,\text{mm}$ . Bandwidth  $\approx 1\%$ – $3\%$ , directivity  $\approx 6$ – $7\,\text{dBi}$ .

**Comparison: Circular vs. Rectangular Patch**

| Property              | Rectangular Patch                                     | Circular Patch                            |
|-----------------------|-------------------------------------------------------|-------------------------------------------|
| Shape                 | Rectangular ( $W \times L$ )                          | Circular (radius $a$ )                    |
| Resonant parameter    | Length $L \approx \lambda / (2\sqrt{\epsilon_{eff}})$ | Radius $a$ from Bessel zero               |
| Dominant mode         | $TM_{100}^x$                                          | $TM_{110}^z$                              |
| Bandwidth             | $\approx 1\%-5\%$                                     | $\approx 1\%-3\%$ (slightly narrower)     |
| Circular Polarization | Possible with square patch + notch                    | Easier — natural symmetry aids CP         |
| Compactness           | Larger footprint                                      | More compact (round shape)                |
| Design Complexity     | Simpler formulas                                      | Slightly more involved (Bessel functions) |

**Circular Patch Design Equations**

The physical radius  $a$  is calculated as:

$$a = \frac{F}{\left\{ 1 + \frac{2h}{\pi\epsilon_r F} \left[ \ln\left(\frac{\pi F}{2h}\right) + 1.7726 \right] \right\}^{1/2}}$$

where:

$$F = \frac{8.791 \times 10^9}{f_r \sqrt{\epsilon_r}} \quad (\text{for } f_r \text{ in Hz})$$

The effective radius  $a_e$  incorporating fringing fields is:

$$a_e = a \left\{ 1 + \frac{2h}{\pi\epsilon_r a} \left[ \ln\left(\frac{\pi a}{2h}\right) + 1.7726 \right] \right\}^{1/2}$$

The resonant frequency for the  $TM_{110}^z$  mode is:

$$(f_r)_{110} = \frac{1.8412v_0}{2\pi a_e \sqrt{\epsilon_r}}$$

## Quick Revision — Module 5

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### Key Formulas and Metrics at a Glance

|                                                            |                                                                                                                              |
|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Rectangular Patch Width (<math>W</math>)</b>            | $W = \frac{v_0}{2f_r} \sqrt{\frac{2}{\epsilon_r + 1}}$                                                                       |
| <b>Effective Dielectric (<math>\epsilon_{reff}</math>)</b> | $\epsilon_{reff} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left[ 1 + 12 \frac{h}{W} \right]^{-1/2}$             |
| <b>Extended Length (<math>\Delta L</math>)</b>             | $\Delta L \approx 0.412h \frac{(\epsilon_{reff} + 0.3)(W/h + 0.264)}{(\epsilon_{reff} - 0.258)(W/h + 0.8)}$                  |
| <b>Physical Length (<math>L</math>)</b>                    | $L = \frac{v_0}{2f_r \sqrt{\epsilon_{reff}}} - 2\Delta L$                                                                    |
| <b>Resonant circular frequency (<math>f_r</math>)</b>      | $(f_r)_{110} = \frac{1.8412v_0}{2\pi a_e \sqrt{\epsilon_r}}$                                                                 |
| <b>Effective circular radius (<math>a_e</math>)</b>        | $a_e = a \left\{ 1 + \frac{2h}{\pi \epsilon_r a} \left[ \ln \left( \frac{\pi a}{2h} \right) + 1.7726 \right] \right\}^{1/2}$ |

### One-Line Definitions

- **Fringing Fields:** The bending of electric field lines at the open-circuit boundaries of a microstrip patch, which enables radiation.
- **Inset Feed:** A feed technique where notch cuts are made in the patch to match the low line impedance ( $\approx 50 \Omega$ ) directly inside the patch.
- **Transmission Line Model:** An analytical model that treats a rectangular patch antenna as two radiating slots separated by a transmission line.
- **Cavity Model:** An analysis model that treats the patch as a cavity bounded by perfect electric and magnetic walls.
- **Full-Wave Analysis:** Direct numerical solvers of Maxwell's equations (e.g., FEM, FDTD) used for complex microstrip geometries.

### Frequently Asked / Viva Questions

**Q1. What is the primary mechanism of radiation in microstrip patch antennas?**

Radiation is generated by the fringing fields at the patch edges. These fringing fields act as slot sources along the radiating edges (separated by  $L \approx \lambda/2$ ).

**Q2. Compare microstrip line feeding and coaxial probe feeding.**

Microstrip line feeds are simple to fabricate but introduce spurious radiation from the feed line, which degrades the radiation pattern. Coaxial probe feeding offers lower spurious radiation and easy matching, but requires drilling holes, making fabrication more complex.

**Q3. What is the purpose of an inset feed in microstrip antennas?**

The input impedance at the edge of a patch is high (typically  $200\text{--}300 \Omega$ ). An inset notch shifts the contact point of the feed line toward the center of the patch,

**Frequently Asked / Viva Questions (contd.)**

where the impedance is lower, matching it to a standard  $50\ \Omega$  transmission line.

**Q4. Why does a microstrip patch antenna have a narrow bandwidth?**

The patch antenna behaves electrically as a high-Q resonant cavity with a small volume. Since the substrate height  $h$  is thin ( $h \ll \lambda$ ), the energy stored in the cavity is high compared to the energy radiated, leading to a narrow bandwidth.

**Q5. Explain the physical meaning of the effective dielectric constant ( $\epsilon_{eff}$ ).**

Because the fringing field lines travel partly through the dielectric substrate and partly through the surrounding air, the wave behaves as if it were propagating through a homogeneous medium with a dielectric constant  $\epsilon_{eff}$  that lies between 1 (air) and  $\epsilon_r$  (substrate).

**Q6. State the advantages of aperture-coupled feeds over direct feeding methods.**

The ground plane separates the feed line from the radiating patch, preventing spurious feed line radiation from distorting the antenna's far-field radiation pattern. It also allows independent optimization of the feed and patch substrate thicknesses.

**Comparison of Patch Feeding Methods**

| Feed Type         | Ease of Fab                    | Spurious Radiation          | Relative Bandwidth |
|-------------------|--------------------------------|-----------------------------|--------------------|
| Microstrip Line   | Very Easy                      | High                        | Narrow (1–2%)      |
| Coaxial Probe     | Moderate (requires drilling)   | Low                         | Narrow (1–3%)      |
| Aperture Coupled  | Complex (multilayer alignment) | Lowest (shielded by ground) | Moderate (4–5%)    |
| Proximity Coupled | Complex (multilayer alignment) | Low                         | Widest (10–13%)    |